PromptShield:
User-AI Information Disclosure and Warning Fatigue

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1 Introduction and Motivation

Can real-time local AI privacy warnings reduce the amount of sensitive information users disclose in AI chat applications without causing excessive warning fatigue?

AI chat systems are becoming a common part of everyday life for personal, academic and professional tasks. Many users may not realize how the information they enter can be stored and processed by third party services. If users are sharing sensitive information such as health, legal, financial or personal information this can become a serious privacy concern. Users can't see how their data is stored or used. Understanding if a real time warning system can help users recognize these risks and make better decisions is important.

We propose a browser extension that works like Grammarly, but for privacy instead of grammar. As a user types into an AI chat window, the extension scans the text in real time and highlights anything that looks sensitive, such as a social security number, phone number, address, or personal details like health or financial information. If sensitive information is detected, the user gets a warning before they send the message. The user can also choose to see an alternative version of their message that keeps the same meaning but removes the sensitive part. Everything runs local in the browser, so no text the user types is ever sent to an outside server for this process.

Our approach targets the exact moment where disclosure happens, when the user is typing, not after the message has already been sent. This gives the warning a real chance to change behavior before any harm is done. Running everything locally also means the tool doesn't create a new privilege risk of its own, this matters because the point of the project is to protect user data. Because our research question is about balancing risk reduction against warning fatigue, our design uses tiered warnings instead of treating every flag the same way. Similar to Grammarly's color-coded underlines, each tier is shown in a different shade: red for high-risk information like SSNs and bank numbers, orange for medium-risk information like home addresses and birth dates, and light blue for lower-risk information like email addresses or a user's city. This lets us test whether a smarter warning system can reduce disclosure without wearing users down.

2 Methods: Data Collection

We plan to recruit at least 4 participants per group member from their own personal network. We'll aim for a mix of ages and backgrounds rather than just classmates. We'll use a short sign up form to confirm we're actually getting some spread in age and AI familiarity, not just repeating the same demographic over each team member.

Each participant will complete a set of short, scripted tasks using a chat interface with our extension installed. The tasks will be designed to naturally invite disclosure of sensitive information, for example, asking AI for help writing an email to a doctor about a symptom or asking help filling out a form that asks for personal details. Some participants will complete one set of tasks with the extension and some with the extension running in the background but not giving them suggestions to avoid releasing sensitive data.

We will collect an anonymized log from the extension showing what type of sensitive information was detected in each task (financial, medical, personal identifier) and whether the participant was in the active suggestion group or the silent background group, without storing the actual text they typed. This log will be collected automatically by the extension during each task, and we'll analyze it directly rather than supplementing it with a post task survey.

We'll be using our own prototype, the browser extension itself, as the main tool being tested, built on the fully in-browser stack. Beyond that, we'll need a mock or sandbox chat interface to run the scripted tasks in, so we're not exposing real AI providers to test data, and a simple way to export and view the extension's anonymized detection logs. We are not using a paper prototype since the extension will be functional enough to test directly by this stage.

For internal validity, we're splitting participants into two groups (one gets active suggestions while the other has the extension running silently in the background) so we can com-

pare disclosure rates between the two conditions rather than relying on self report. We'll also keep the scripted tasks identical across both groups so any difference we see comes from the extension's behavior, not from the task content. For external validity, we know our sample is small and drawn from personal networks, but recruiting across different ages and backgrounds helps us catch whether the effect holds beyond just students our own age, even if we can't fully generalize to all AI users. For construct validity, we're using a clear, fixed set of sensitive information categories that matches what our detection model actually flags, so sensitive disclosure is measured consistently across both groups rather than left open to interpretation.

Since tasks are designed to invite disclosure of sensitive sounding information. We'll have participants use fictional placeholder details instead of anything real, and this will be stated directly in the task instructions. Well get informed consent from every participant before starting and explain they can stop at any time. Since we're not sorting the actual text participants type, only anonymized category logs, the privacy risk to participants is low.

3 Methods: Data Analysis

Our analysis will primarily be quantitative. The main outcome measured is the rate of sensitive information disclosure under the two experimental conditions. Because the extension records the category of sensitive information detected without storing the actual text, we can compare how often participants attempt to include sensitive information and how often that information is actually sent in each condition.

For each participant, we will calculate the number and percentage of scripted tasks in which sensitive information is detected. For participants in the active-warning condition, we will also record whether the warning changes their behavior, such as removing or replacing the flagged information before submission. We will compare these disclosure rates against the silent-background control group to estimate whether the warnings reduce sensitive disclosures.

We will also analyze warning fatigue by measuring how many warnings each participant receives and whether the rate at which participants respond to warnings changes over the course of the study. We will also separate results by warning severity and sensitive-information category, such as personal identifiers, medical information, and financial information to determine whether participants respond differently depending on the type or risk level of the disclosure.

To ensure analysis quality, the experimental tasks, sensitive-information categories, and logging rules will be defined before participant testing begins and will remain consistent across both conditions. We will test the extension on a labeled set of example prompts before the user study to measure detection precision and recall and identify obvious false positives or false negatives. We will also inspect the anonymized logs after collection for missing or inconsistent records before performing the final analysis.

4 Anticipated Challenges and Limitations

Several challenges may affect the performance of the extension and the interpretation of our results. One concern is false positives, where harmless text is incorrectly flagged as sensitive. To mitigate this, we will evaluate the system's precision using a labeled test set before participant testing and allow users to easily dismiss warnings. Matches that are less certain will also receive less intrusive warnings so that ambiguous cases do not interrupt users unnecessarily.

False negatives are another limitation, particularly when sensitive information is expressed indirectly or spread across multiple prompts. We will measure recall using the same labeled test set and refine our detection rules before beginning the user study. However, we recognize that no detection system will identify every possible privacy risk.

Context also presents an important challenge because whether information is sensitive can depend heavily on how and where it is used. A local detection system may be able to identify explicit information such as financial details, medical information, or personal identifiers, but it may struggle with information that only becomes sensitive when interpreted in context. For this reason, our evaluation will focus primarily on clearly defined categories of sensitive information rather than attempting to capture every possible form of privacy risk.

Warning fatigue may also reduce the effectiveness of the system. If users receive too many alerts, they may begin dismissing them automatically or ignoring future warnings. Our tiered warning design attempts to reduce this problem by reserving the most noticeable warnings for higher-risk information while presenting lower-risk detections less intrusively. During testing, we will track the number of warnings generated per task and examine whether users become less responsive to warnings over time.

Finally, our relatively small sample size limits how broadly the results can be generalized. With approximately four participants recruited by each team member, the study may not represent the full population of AI users. We will attempt to improve the diversity of the sample by recruiting participants with different ages and levels of familiarity with AI systems and by randomly assigning participants to experimental conditions. Even with these measures, we will treat the results as preliminary and clearly acknowledge the limited sample size when interpreting our findings.

5 Related Works

Prior research has established that users regularly disclose sensitive information when interacting with large language models. Mireshghallah et al. analyzed naturally occurring

human-LLM conversations and found that personally identifiable information and other sensitive content appear frequently, including in contexts where disclosure may not be necessary [2]. Their findings motivate the need for interventions that occur before sensitive information is submitted. Similarly, Zhang et al. combined analysis of real ChatGPT conversations with interviews of 19 users and found that users often have incomplete mental models of how conversational AI systems handle their data. They also observed that human-like interaction can encourage increasing levels of disclosure [5]. Together, these studies demonstrate the prevalence of the problem, but they primarily characterize disclosure behavior rather than directly testing preventative mechanisms.

Research on security warnings also informs the design of privacy interventions. Akhawe and Felt conducted a large-scale field study of browser security warnings and showed that user responses depend substantially on warning design and context [1]. Users ignored or bypassed some warning types much more frequently than others, demonstrating that repeatedly presenting intrusive warnings can reduce their effectiveness. This motivates our use of tiered warnings rather than treating every detected disclosure as equally severe.

More recent work has explored automated privacy protection specifically for LLM interactions. Zhou et al. introduced Rescriber, a browser extension that detects personal information in prompts and allows users to redact or abstract it before submission [6]. In a 12-participant study, Rescriber reduced unnecessary disclosure, and participants rated a smaller Llama3-8B-based implementation comparably to a GPT-4o-based version. This work is closely related to our proposed system because it demonstrates the feasibility of user-facing, on-device data minimization. Our project differs by focusing specifically on whether real-time, risk-tiered warnings change disclosure behavior and whether those warnings introduce measurable warning fatigue.

Serenari and Lee proposed LOPSIDED, a privacy agent that detects personally identifiable information and replaces it with contextually appropriate pseudonyms before sending a prompt to a remote LLM [3]. The system then restores the original entities in the generated response. This approach protects sensitive information while attempting to preserve semantic utility, but it operates largely as an automated mediation layer rather than studying how visible warnings affect user decision-making. Our system instead keeps users directly involved in deciding whether and how sensitive information should be modified.

PrivacyLens, introduced by Shao et al., examines a related problem from the model side by evaluating whether language-model agents follow contextual privacy norms when taking actions [4]. Their experiments demonstrate a gap between models' ability to identify privacy concerns abstractly and their behavior when executing tasks, with sensitive information still being leaked in agent scenarios. PrivacyLens therefore suggests that privacy protection should not rely solely on an LLM recognizing privacy risks internally. Our work addresses a complementary point in the interaction by attempting to prevent sensitive information from reaching the remote model in the first place.

Overall, prior work establishes both that sensitive disclosure in conversational AI is common and that automated privacy mediation is technically feasible. Our project builds on these findings by studying the human factors of the intervention itself. Rather than only detecting or automatically sanitizing sensitive information, we use severity-based warnings and allow users to decide how to respond. Our primary contribution is therefore an evaluation of whether a fully local, real-time privacy warning system can reduce sensitive disclosure while avoiding excessive warning fatigue.

6 Proposed Budget

Our proposed budget is $100. A small portion of the budget will cover the $5 Chrome Web Store developer registration fee needed to publish and distribute the browser extension. The remaining funds will be used to provide $5 gift cards to study participants as compensation for their time.

7 Timeline

Table 1: Project Timeline

Date	Milestone
Sept. 27	Project proposal report due.
Oct. 4	Test candidate local LLMs and select a base model for sensitive-information detection.
Oct. 10	Generate the fine-tuning dataset with labeled financial, medical, and personal identifier examples across three risk tiers.
Oct. 17	Fine-tune the model and evaluate precision and recall on a held-out test set.
Oct. 20–22	Project progress presentation.
Oct. 31	Implement the Chrome extension and mock chat interface; finalize personas, task scripts, and consent forms.
Nov. 7	Conduct the user study, with each team member recruiting at least four participants randomly assigned to active or silent conditions.
Nov. 14	Analyze logs, compare disclosure and warning rates between groups, and finalize the project report.
Nov. 17–24	Final project presentation.

Table 2: Approximate Team Responsibilities

Team Member	Responsibilities
Hamza	LLM testing and selection, fine-tuning, and precision/recall evaluation.
Aadya	Building and labeling the fine-tuning dataset and held-out test set.
Ruhi	Chrome extension development, mock chat interface, and logging system.
Karim	Study design, including personas, task scripts, consent forms, and log analysis.

References

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